

Response to reviewer and editor comments

Manuscript: “A real-time reservoir inflow forecast evaluation framework” (MS 8299185), submitted to the *Journal of the American Water Resources Association*.

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We thank the editor and the four reviewers for their careful and constructive comments. All reviewers agreed on one point, which the handling editor also identified: the manuscript described a framework without stating plainly what was new about it. We agree. The original draft presented the five-step process as though the process itself were the contribution, when the actual contribution is narrower and more useful than that. We have restructured the introduction and methods around an explicit statement of contribution, added a section comparing the framework to retrospective verification practice, and added a case in which a tailored metric and the generic metrics support different conclusions about the same forecast.

We have also substantially expanded the description of the study system and the forecast products, added a smoothing sensitivity analysis, characterized the hydrology of the evaluation period against the long-term record, cited the forecast value literature, and corrected a number of errors in the methods section that the reviewers identified.

We would like to note two corrections to statements in the original manuscript:

- The evaluation period was stated incorrectly. The manuscript gave the period as July 2023 to May 2025. The record actually analyzed runs from June 2023 to April 2025.
- The forecast products do not cover the forecast period uniformly. The commercial forecast labeled B spans the full period, while forecast A and the in-house forecast both end in November 2024. The metrics in Table 1 are therefore computed over overlapping but not identical sample sizes. We have reported the sample size for each product in the table, verified that restricting the comparison to the 510 days on which all three products are present does not change the ranking at any location

Reviewer comments are set in indented italics; our responses follow in plain text. New or revised manuscript text is set indented in roman type.

Editor comments

The greatest barrier, from my reading of the manuscript and review of the comments, is the question of novelty and reproducibility. Reviewers 1, 3, and 4 note that there is no distinct statement of novelty, with reviewer 3 particularly noting that the manuscript seems to suggest that the framework is the novelty without fully describing the novelty.

Agreed, this was an oversight. We have added a statement of contribution to the end of the introduction:

“The contribution of this work is a specific operating procedure for the case where a utility must choose between competing forecast products under live operating conditions and cannot

run a hindcast. That procedure has two components that distinguish it from retrospective verification practice. First, the operators participate in metric selection and design and can modify, remove, or append metrics to the evaluation as more data is collected. Second, the evaluation runs on the same operational data the operators use which reflects the real day-to-day operational decisions.”

We have also rewritten the paragraph preceding this one so that the gap being addressed is stated directly. Traditional verification requires hindcasts over a climatologically representative period, which is expensive and often infeasible; the fallback of evaluating archived operational forecasts fits poorly with products that are updated continuously, trained on evolving data streams, hand adjusted by operators, or developed by third parties. That is the gap.

The claim that operator participation changes the outcome is supported by a specific result rather than asserted. The forebay compliance exceedance metric, which exists in the analysis only because the operators asked for it, separates the two commercial products where the generic metrics do not: from a neutral forebay start the products are difficult to distinguish, with all exceedance probabilities below 10%, whereas from the non-neutral start that operators are routinely in, forecast A reaches 23.7% at Bellows Falls against 1.8% for forecast B. This is now presented as a result in Section 4.4 rather than left implicit.

On reproducibility, the evaluation code is public (<https://github.com/HydroWIREs-PNNL/in-flow-forecast-evaluation>) and sample data are on Zenodo (<https://doi.org/10.5281/zenodo.16921728>). We have added the parameter values that were previously left implicit, most importantly the smoothing window (Reviewer 4, comment 4.1), together with a sensitivity analysis over that parameter; the drainage areas and USGS reference gages for all six evaluation points (Reviewer 1, comment 1.5); the issue frequency and horizon of each product; and the per-product sample sizes. The proprietary forecast time series cannot be released, which we now state as a limitation alongside what a reader would need to replicate the analysis on their own system.

Reviewer 1

The topic and the contents are relevant, and it is very interest for hydropower operators and decision-makers. However, the novelty and main contributions of this study are unclear.

Thank you for the review. The comments on framing, the missing system description, and the table and figure quality were all warranted and we address each below. Your comment 1.6, asking us to check an apparent anomaly in the NSE values, turned out to identify something genuinely informative rather than an error, and it has become a result in the revised manuscript.

1.1 Novelty and main contributions

The novelty and main contributions of this study are unclear. Please explain and discuss in detail.

We agree this was missing. Please see our response to the editor above for the new statement of contribution. In brief, the contribution is a procedure for the specific situation in which a utility must choose among competing forecast products under live operating conditions and a hindcast is not available. What distinguishes it from standard verification is that metric selection is deferred and driven by operator input rather than fixed in advance, and that the evaluation runs on the

operational data feed rather than an archived dataset.

We have also added a paragraph to the discussion on why this matters now. Forecast products using data driven methods are being marketed to utilities at an increasing rate, and the verification statistics vendors supply are computed on their own hindcasts against their own choice of reference data. A utility generally has no standard procedure for checking those claims against its own system, and as we now report, it often cannot inspect the model at all. This framework is intended to be that procedure.

1.2 Persistence as a benchmark at long lead times

Lines 70–72: “‘persistence’ forecasts which use the last observed value carried out for all future timesteps to get a sense of the arguably worst possible forecast. Persistence is a competitive benchmark for short term operations, from hours to 2-day horizons”. However, Fig. 3 and Fig. 4 show that 10-day and months “persistence forecasts” as benchmark which is unrealistic in practice.

You are correct, and this is an internal inconsistency in the original manuscript: we stated that persistence is competitive only to about two days and then plotted it out to ten, while also noting that a climatological benchmark would be more appropriate at longer horizons without actually using one. Reviewer 2 raised a related point about the description of persistence in their comment 2.7.

We have revised the benchmark discussion so that the horizon over which persistence is meaningful is stated explicitly, and so that its appearance at longer lead times in the figures is qualified rather than presented as a reference:

“Persistence is a simple but competitive benchmark at short lead times, from hours to roughly two days, and is the appropriate reference for day-ahead operations, which is the horizon of interest for the system evaluated here. It degrades quickly beyond that horizon, where a climatological forecast is the more appropriate lower bound. Evaluators should match the benchmark to the horizon of interest: persistence for hourly to 2-day operations, climatology for weekly to seasonal horizons. Where persistence is plotted at longer lead times in the figures below, it is retained only to show the scale of degradation and should not be read as a meaningful reference at those horizons.”

The figure captions now carry the same qualification, so a reader looking only at Figures 3 and 4 is not misled.

We should be straightforward about what we have not done. We have not added a climatological benchmark to the figures. Constructing a defensible daily climatology for these locations requires a long record of calculated inflow at the dams, and the calculated inflow series available to us begins with the evaluation. We could substitute a climatology derived from the nearby USGS gage record, but the gage record and the calculated inflow series differ in drainage area and in regulation, so the resulting benchmark would not be commensurable with the quantity being verified, and a benchmark that is not commensurable is worse than none. We have therefore stated the recommendation, which is general and which we stand behind, and been explicit that the day-ahead horizon is the operationally relevant one for this system and is where persistence is the correct reference. We would rather make that limitation visible than add a benchmark we cannot defend.

1.3 Operating problems and the forecasting scheme in practice

What are the main operation problems of these hydropower stations? Which forecasting scheme is used in practice?

This context was missing and it is needed to make the tailored metrics comprehensible. We have added a subsection, “Operating context,” which states the following. The three dams have limited storage relative to their inflow and are operated as run-of-river facilities. Their daily cycle consists of an upward ramp, a period of flexible generation, a downward ramp, and a refill period. Generation is bid into the ISO New England day-ahead market, so the day-ahead inflow volume is the quantity that matters most, since it sets what the operators can commit to. The binding operational constraint comes from the FERC relicensing of the three lower Connecticut River dams, which imposed a ± 0.5 foot (≈ 15 cm) limit on forebay fluctuation, with penalties for exceedance. The central operating problem is therefore committing generation a day ahead on the basis of an inflow forecast without violating the forebay band, and this is what motivated the cumulative-error and threshold-exceedance metrics.

On the forecasting scheme used in practice: at the time of the evaluation the operators used an in-house forecast built from National Weather Service River Forecast Center forecasts, routed to the dam locations and adjusted using operator judgment. This is the incumbent product against which the two commercial forecasts were evaluated, and it is labeled GRH in the results. It extends only about 32 hours, because the day-ahead bid is the only decision it is used for.

1.4 Description of the forecast products

Two commercial forecasts (A, B) and the GRH in-house forecast (GRH) should be described in detail, such as what type of hydrologic model used? How about the calibration and validation results, what type input data requirement etc.

Reviewers 2 and 4 raised the same point (comments 2.16 and 4.3). We agree that anonymization without any methodological characterization leaves readers unable to interpret the results, and we have added what we are able to disclose:

“Forecast A is produced by a consulting firm using a conceptual rainfall-runoff model with hydraulic routing through the Connecticut River mainstem, driven by a commercial numerical weather prediction vendor, and issued once daily at hourly timestep out to roughly 8 days. Forecast B is produced by a technology firm using a machine learning model trained on gauge and remotely sensed catchment observations, issued twice daily at hourly timestep out to 10 days. The GRH in-house forecast combines RFC forecasts with routing and operator judgment and extends only through the day-ahead period, roughly 32 hours, because that is the horizon it is used for. Neither vendor released model internals or calibration and validation diagnostics, and we did not have access to them. That limitation is itself a finding: a utility evaluating commercial forecast products generally cannot inspect them and must judge them on operational performance, which is part of the motivation for this framework.”

The issue frequencies and horizons above are taken from the forecast archive itself rather than from vendor documentation, so they describe what was actually delivered.

On calibration and validation results specifically, we are not able to provide them. The agreements under which the products were supplied prevent us from naming the vendors or describing model internals, and the vendors did not release calibration diagnostics to us. We recognize this is unsatisfying from a scientific standpoint. We have chosen to state the gap plainly and treat it as

a finding about the procurement situation operators face, rather than leaving readers to assume the information was available and simply omitted. We have also added that a full procurement evaluation should weigh operational cost, data dependencies, and ease of integration alongside skill, since these differ substantially between the two commercial products and are part of what a utility is actually choosing between.

1.5 Watershed information

Three main hydropower producing dams in the system, Wilder, Bellows Falls, and Vernon (Figure 2) as well as three main tributaries, the Ottauquechee, Sugar and White Rivers. The watershed information, such as area, river length etc. should be described in detail.

Agreed, this was a gap. We have added a table of physical characteristics for all six evaluation points, with drainage areas taken from the USGS National Water Information System for the reference gage at each location:

Location	Type	USGS reference gage	Area (mi ²)	Area (km ²)
Wilder	Mainstem dam	01144500 West Lebanon, NH	4,092	10,598
Bellows Falls	Mainstem dam	01154500 North Walpole, NH	5,493	14,227
Vernon	Mainstem dam	01156500 Vernon, VT	6,266	16,229
White River	Tributary	01144000 West Hartford, VT	690	1,787
Sugar River	Tributary	01152500 West Claremont, NH	269	697
Ottawaquechee River	Tributary	01151500 North Hartland, VT	221	572

We have also added text describing the basin. The mainstem drainage areas are nested, so the area above Vernon includes those above Bellows Falls and Wilder. This nesting matters for interpreting the results and explains the downstream propagation of bias noted in the original manuscript, which we now quantify: for forecast A, day-ahead PBIAS at Bellows Falls is 0.4% and grows to −19.1% by day 7, and the growth is larger at the downstream locations. Because releases from each dam become inflow to the next, an error in an upstream forecast enters the downstream forecast as an error in its boundary condition and compounds with lead time.

Basin hydrology is snowmelt influenced, with a spring freshet typically spanning March through May, convective rainfall in summer, and rain and rain-on-snow driven events in winter. Several tributaries have upstream flood control regulation by the U.S. Army Corps of Engineers, which contributes to inflow variability that is not directly weather driven. We have not added river lengths, as the drainage areas and the nesting relationship are what the results depend on and we did not want to pad the table with figures that play no role in the analysis.

1.6 NSE values in Table 1

In Table 1, the NSE values of forecast scheme A at Wilder, Bellows and Vernon watersheds are 0.72, 0.55, 0.81, respectively. Please check the evaluation results. The forecasting accuracy at the Bellows might high than that at the Wilder.

Thank you for checking this. We have re-run the calculation from the processed data and the values are correct as published: for forecast A, day-ahead NSE is 0.72 at Wilder, 0.55 at Bellows Falls, and 0.81 at Vernon. Your question prompted us to look into why, and the answer turned out to be a genuine finding about forecast A rather than an artifact, so we have added an explanation rather

than simply confirming the numbers.

Two separate effects are at work and it is worth separating them.

The first is that NSE and KGE are normalized by the variance of the observations at each location, so their values are not directly comparable across locations with different inflow variability (Knoben et al., 2019). The persistence benchmark makes this visible: persistence alone achieves NSE of -0.81 at Wilder, 0.39 at Bellows Falls, and 0.80 at Vernon. Flows become progressively more regulated and therefore smoother downstream, so at Vernon persistence explains most of the variance and little room is left for a forecast to improve on it. This is now explained in the text, and we have added a skill score column to Table 1 computed against the persistence forecast so that readers have a normalization-free comparison available.

The second effect, and the one that actually answers your question, is that forecast A really is weakest at Bellows Falls, and the reason is visible in the KGE decomposition. For forecast A, the variability and bias ratios at Bellows Falls are essentially perfect ($\sigma_f/\sigma_o = 1.00$, PBIAS = 0.4%), so the entire KGE shortfall there comes from correlation, which is 0.77 against 0.88 at Wilder and 0.91 at Vernon. In other words, forecast A gets the magnitude and the average volume right at Bellows Falls but gets the timing wrong.

What distinguishes Bellows Falls is that it is the only evaluation point whose inflow includes substantial unregulated local runoff, from the Ottawaquechee, Sugar, and White Rivers, which enter between Wilder and Bellows Falls. Evaluating the three tributaries individually shows this is where forecast A struggles: day-ahead NSE is -0.02 on the Ottawaquechee, 0.08 on the Sugar, and -1.22 on the White, against 0.50 , 0.78 , and 0.77 respectively for forecast B. Computing the local inflow directly as the difference between Bellows Falls and Wilder gives NSE of 0.07 and correlation of 0.58 for forecast A, against 0.84 and 0.92 for forecast B. Forecast A therefore has very little skill in the unregulated tributary reach, and Bellows Falls is the mainstem location where that weakness shows up most strongly because that is where the tributaries enter.

So the answer to your question has two parts. Part of the cross-site pattern is a normalization artifact and should not be read as a statement about forecast quality. But the low value at Bellows Falls specifically is real, it is a timing rather than a magnitude problem, and it traces to poor performance on the unregulated tributaries. We have added the tributary metrics to the manuscript and the following explanation:

“NSE and KGE are normalized by the variance of the observations, so values are not directly comparable across locations with different inflow variability. Comparing raw values across sites can therefore give a misleading ranking, and we recommend reporting skill scores relative to a benchmark alongside the raw metrics in any multi-site evaluation. Beyond that normalization effect, the KGE decomposition is diagnostic here: forecast A’s shortfall at Bellows Falls is almost entirely a correlation deficit, with essentially no variability or bias error, indicating a timing problem rather than a magnitude problem. Bellows Falls is the only mainstem evaluation point whose inflow includes substantial unregulated local runoff, and forecast A performs poorly on those tributaries individually, which accounts for the pattern.”

This is a good illustration of why we recommend KGE over NSE, since the decomposition localized the deficiency in a way a single aggregate score could not, and we are grateful the question was raised.

1.7 Table format

The format of Tables in the manuscript are poorly designed and needed be changed.

Agreed. The tables were generated with default settings and not revised. Table 1 has been rebuilt with `booktabs` rules and no vertical rules, with rows grouped by location and the location named once per group, numeric columns aligned, consistent significant figures across metrics, and units in the column headers. It has gained three columns: the number of day-ahead forecasts evaluated for each product, which is necessary because the products cover different portions of the period; RMSE in cfs, requested by Reviewer 4 in comment 4.7; and a KGE skill score computed against the persistence forecast, described in comment 1.6 above. Two new tables have been added, one giving the basin characteristics (comment 1.5) and one giving the smoothing sensitivity analysis (Reviewer 4, comment 4.1), both in the same style.

1.8 Figures

The figures also need be redrawn.

We have revised the figure captions substantially and we describe below what we have and have not changed in the figures themselves, since we would rather be precise than claim a wholesale redraw.

Captions are now self-contained. Every caption states the quantity plotted, the meaning of any reference line, and the units. The Figure 3 and 4 captions now state that the no-skill threshold is NSE of 0 and KGE of -0.41 , rather than referring vaguely to “no skill over the mean,” and note that persistence is only a meaningful reference over the first two days (comment 1.2). The Figure 5 caption now specifies that the error bars span the 2.5th to 97.5th percentile, which Reviewer 4 flagged in comment 4.8, and states the pass-inflow assumption and the compliance band. The Figure 1 caption states the smoothing window and gives the SI conversion for cfs, which Reviewer 2 requested in comment 2.17. The Figure 6 caption now explains why the lower panel is the operationally relevant case.

On the figures themselves, we would welcome more specific guidance. We are willing to change the palette, line styles, text sizes, and output format, and we agree these could be improved, particularly the legibility of axis and legend text at print size and the consistency of the color assignment across panels. We have not yet regenerated them because we did not want to redraw them once on our own guess at the objection and then again after learning what it was. If the reviewer would indicate which aspects they consider deficient, we will address them directly in the next round.

Reviewer 2

The core work is solid and the practical contribution is genuine. Assuming the major comments are addressed I recommend this manuscript is accepted with minor revision.

Thank you for a detailed and technically careful review. Several of the minor comments identify outright errors in the methods section, including one in an equation, and we are grateful they were caught before publication. The two major comments, on engaging the forecast value literature and on positioning the contribution, have both changed the manuscript substantially.

2.1 Engagement with the forecast value literature

The stated goal of the framework is to help operators evaluate whether a forecast product will improve operations enough to justify investment. The paper does not include any assessment of forecast value and the authors acknowledge this gap (L172–177), framing it as “an important next step.” I appreciate this is a difficult problem, but the paper needs to at least engage with the existing forecast value literature and discuss how the presented metrics relate to the investment question. [...] The authors do not need to perform a full value assessment, but they should reference this literature, discuss how it connects to their framework, and provide operators with at least a conceptual pathway from the metrics presented to the investment decision the paper is motivated by.

This is a fair criticism and it identifies a genuine gap between what the paper promises in the introduction and what it delivers. We motivate the work with the investment question, present only forecast quality metrics, and defer value to future work. Engaging the literature was the missing step.

We have added a discussion subsection, “From forecast quality to forecast value,” which draws the quality–value distinction and gives operators a pathway from the metrics presented to the investment decision:

“The metrics presented here are measures of forecast quality, not forecast value. Quality is a statistical property of a forecast paired with observational data. Value depends additionally on human decisions and the cost associated with forecast errors. A forecast can be more statistically accurate and yet carry less value, if the additional accuracy falls in a range where the operator would not act differently. We refer the reader to Farkas et al. (2025) for a comprehensive review of the economic value of forecasts.

The tailored metrics in Section 4.4 are a step towards assessing forecast value. For example, the forebay exceedance probability is a quantity with a direct financial consequence, since exceedance incurs a penalty under the FERC license. This is a partial valuation, since it captures an upper limit on compliance risk but it is not put in terms of generation revenue. A more complete assessment of revenue requires a decision-support model which includes reservoir operations and market conditions. We recommend that operators treat a real-time quality evaluation as the first of two stages. The evaluation establishes which forecast products are plausibly worth adopting and produces the archive of forecasts that may be used to drive the second step where value is assessed.”

For the literature we have cited Farkas et al. (2025), a recent systematic review of the economic value of weather forecasts, in preference to citing individual methods. We considered writing a more detailed treatment of specific value metrics, including the relative utility value approach, and drafted one. On reflection we judged that a paragraph comparing value metrics we do not apply would read as padding in a paper whose subject is quality evaluation, and that a pointer to a comprehensive review serves a practitioner reader better. If you or the editor would prefer the more detailed treatment, we are glad to restore it.

We have not attempted a value assessment in this revision. Doing so credibly requires the utility’s damage function and a representation of how operators respond to the forecast, which returns to the scheduling model coupling discussed under comment 4.9, and we did not want to present a value estimate we could not defend.

2.2 Positioning of the contribution

The five-step iterative process (collect data, compute metrics, visualise, solicit feedback, iterate) reads more as standard good operational practice than a novel framework contribution. I feel that the tailored metrics and operator feedback loop are where the genuine contribution lies, especially in the real-time context, and these deserve more prominence and deeper treatment.

We agree, and your diagnosis matches Reviewer 3’s and the handling editor’s. Yours is the most useful of the three because it says not only what is missing but where the contribution actually is, and we have restructured the paper accordingly. Please see our response to the editor for the new statement of contribution.

The specific changes that follow your recommendation. Steps 1 through 3 are now explicitly labeled as generic practice for which we claim no novelty. Soliciting feedback and acting on it have been separated into two distinct steps, “Gather feedback” and “Update metrics,” since collecting operator comment and actually changing the metric set are different activities and the second is the one that matters; the process is now described in six steps rather than five. The mechanics of both are documented rather than asserted, including a table tracing each tailored metric to the operator concern that prompted it and the approximate month it was added (see comment 3.2). The two timescales conflated in the iteration step, metric convergence and record length, are now separated (comment 3.3). We have added the case where a tailored metric separates two products that the generic metrics do not (comment 3.5), and a subsection comparing the framework against conventional retrospective evaluation (comment 3.4). The real-time context is now stated as part of the contribution rather than as a constraint we worked around, and we have added that operational reliability is part of what the evaluation measures: a forecast that cannot be produced operationally should not be considered in a real-time evaluation at all.

2.3 Hyphenation of “real time”

(L2, Title) “real time” should (perhaps) be hyphenated as “real-time” when used as an adjective (also applies throughout the manuscript, e.g. L44, L48, L50, L110, L116).

Corrected throughout, including the title, which is now “A real-time reservoir inflow forecast evaluation framework.” Reviewer 4 raised the same point in comment 4.5. The manuscript contained 29 instances of “real time” and none of the hyphenated form, so the inconsistency was in fact between the manuscript and the editor’s own rendering of the title. We reviewed each instance individually and hyphenated in attributive position (“real-time evaluation,” “real-time data”), leaving the unhyphenated form where it functions as an adverbial phrase (“conducted in real time,” “can be used in real time”).

2.4 Verification improving alongside forecasts

(L12–13) The sentences seem to imply that we should expect verification methods to improve alongside improvements in forecasts, but this isn’t the case. Also, the first sentence is about forecast and scheduling methods but the second sentence links it to operational paradigms somewhat abruptly. Suggest restructuring so the research gap is clearer.

You are right on both counts, and the implication you identify was not one we intended. Verification methodology has not tracked forecasting methodology, which is closer to the gap we meant to describe. We have rewritten the passage so that the gap is the subject rather than an inference the reader has to draw:

“The quality of hydrologic forecasts and hydropower scheduling has rapidly improved in recent years due to the emergence of new data driven approaches and scheduling optimization. Verification practice has not changed correspondingly. The metrics in common use predate these developments and are typically applied retrospectively to archived forecasts, an approach that fits poorly with products that are updated continuously, trained on evolving data streams, and delivered to operators as services rather than as models they can inspect. The gap this study addresses is therefore not a shortage of metrics but the absence of a procedure by which an operator can evaluate such a product on their own system under operating conditions.”

The phrase “rather than as models they can inspect” is now supported later in the paper, where we report that neither vendor released model internals or calibration diagnostics.

2.5 Dismissal of ensemble metrics

(L42–44) The distinction between deterministic and probabilistic is important and well noted. However, the dismissal of ensemble evaluation metrics seems premature given that the authors later suggest probabilistic forecasts may help (L170). Consider briefly noting what ensemble metrics could be added if ensemble forecasts were available.

Agreed, and the inconsistency is a fair catch: we set probabilistic evaluation aside early and then invoked probabilistic forecasts as a remedy later. Reviewer 4 asked for the same addition in comment 4.4, so we have added a methods subsection, “Extension to probabilistic and ensemble forecasts,” covering CRPS, rank histograms, and reliability diagrams. Please see our response to comment 4.4 for the detail.

We have also softened the framing at L42–44 so that the focus on deterministic forecasts is presented as a property of the products under evaluation rather than a judgment about ensemble metrics. The revised text notes that all three products evaluated here are deterministic, which is why the metrics we describe are deterministic, and points forward to the new subsection.

One connection is worth drawing out, since it strengthens the point you make. The forebay exceedance metric we develop is already a probabilistic statement derived from a deterministic forecast, computed across the empirical distribution of that forecast’s past errors. Given an ensemble it would be computed directly from the members, which is both simpler and better founded. So the tailored metric and the ensemble metrics are not separate directions; the former is what one does in the absence of the latter.

2.6 The argument against hindcast evaluation

(L45–47) The argument against hindcast evaluation is reasonable but somewhat overstated. Hindcast evaluations do not necessarily require operators to re-operate the system, especially if verifying forecasts of environmental variables such as inflows, which you are. Suggest softening this claim or explaining how operators are in the loop for the inflow forecasts. The first major comment above may be relevant here.

You are correct and the claim as written was overstated. Verifying an inflow forecast against observed inflow does not require re-operating the system, and we should not have implied otherwise. The re-operation requirement applies to the value assessment rather than to the quality assessment, which is the distinction your first major comment prompted us to draw. We have rewritten the passage accordingly:

“A hindcast evaluation of forecast quality does not itself require re-operating the system, since forecasts of inflow can be compared directly against observed or calculated inflow. Two complications nonetheless arise in this setting. First, where the forecast involves human judgment, as the in-house forecast evaluated here does, a hindcast requires forecasters to reconstruct the decisions they would have made under past conditions. Those conditions include the wholesale electricity prices the operator was bidding against, which drive generation scheduling and therefore release decisions, along with the state of the data feeds at the time. Reconstructing these after the fact is difficult and the reconstruction is difficult to validate. Second, extending the evaluation from forecast quality to forecast value does require an operations model or renewed operator engagement, because the value of a forecast depends on the decisions taken in response to it. A real-time evaluation avoids the first complication and generates the forecast archive that the second requires.”

There is a further dependency in our case that is worth stating, and which we had not made explicit. The calculated inflows we verify against are derived from forebay elevation changes and dam outflow, so the observation record is itself a function of how the system was operated. This does not require re-operation to interpret, but it does mean our reference data is not independent of operations in the way a gauge record would be. We have added this to the limitations.

2.7 Persistence as “arguably worst possible forecast”

(L71) Describing persistence as “arguably worst possible forecast” is incorrect. Persistence is a competitive benchmark for short lead times as the authors themselves note in the next sentence. Climatology is typically considered the lower bound for longer horizons. Suggest rephrasing to “a simple but competitive benchmark.”

Corrected, and you are right that the original phrasing contradicted the following sentence in our own text. We have adopted your suggested wording. Reviewer 1 raised the related point about plotting persistence at long lead times in comment 1.2; please see that response, which includes the revised passage and an explanation of why we have not added a climatological benchmark to the figures despite recommending one.

Our own results make your point concretely: persistence achieves day-ahead NSE of 0.80 at Vernon, which is competitive with the commercial products, and it is precisely this that reveals forecast A adds almost nothing there.

2.8 Smoothing window width

(L80–81) A triangular smoother is reasonable but the choice of window width is not discussed. The smoothing window will affect the comparison with forecasts and should be selected with care. Please note the window used and perhaps the sensitivity to this choice on your findings.

Reviewer 4 raised this as their first major comment and we have addressed it with both the value and a sensitivity analysis. The window is 24 hours, implemented as two successive 24-hour moving averages of the hourly calculated inflow, which is what produces the triangular weighting. It was chosen to match the day-ahead operational timescale. Metrics improve modestly and almost monotonically with window width, the largest change being 0.05 NSE, but the ranking of the three products is unchanged at every window width at every location. Please see our response to comment 4.1 for the full table and the accompanying caution.

2.9 KGE name and citation

(L92) “King-Gupta Efficiency” should be “Kling-Gupta Efficiency.” Also, “(refs)” is a placeholder that needs to be replaced with actual references (Gupta et al. 2009 presumably).

Both corrected, with Gupta et al. (2009) as you surmised, and Nash and Sutcliffe (1970) added for NSE in the same sentence. Reviewer 4 flagged the same two items in comment 4.6. Thank you for catching the misspelling.

2.10 The claim about KGE and large values

(L92–93) The claim that KGE is not skewed by large values while NSE and RMSE are requires clarification. KGE and NSE both use correlation as a component. The relevant difference is that KGE decomposes performance into correlation, variability ratio, and bias ratio rather than using squared error. The statement as written is misleading.

You are right, and your characterization is the correct one. The original sentence attributed the difference to the squared error term, but KGE also incorporates correlation, which is itself computed from squared deviations, so the stated reason does not hold. We have replaced it with your framing:

“We recommend KGE because it decomposes forecast performance into correlation, a variability ratio, and a bias ratio, so that a low score can be attributed to a specific deficiency rather than only registered as large aggregate error. NSE and RMSE aggregate squared error into a single number, which makes them more strongly influenced by the largest errors and less diagnostic about their source.”

This correction also improves the paper’s internal consistency, since the decomposition argument is what we rely on elsewhere: forecast A’s poor performance at Bellows Falls shows up as a variability and correlation problem rather than a bias problem, which is visible in KGE and not in NSE.

2.11 Placement of the Knoben et al. citation

(L95–96) The KGE threshold of -0.41 is attributed to comparing against the mean of observations but the Knoben et al. (2019) citation is on L97, not adjacent to the claim. Consider restructuring so the citation directly supports the threshold statement.

Corrected. The citation now appears with the threshold statement it supports, and we have made the logical connection to the following sentence explicit, so that the reason NSE and KGE are not interchangeable is stated rather than merely asserted:

“Values of KGE that are greater than -0.41 are better than using the mean of the observations as a predictor (Knoben et al., 2019), but larger values are preferred up to the maximum value of 1 for a perfect forecast. Because this threshold is not zero, NSE and KGE have different properties and should not be used interchangeably, and intuition built on NSE values does not transfer to KGE.”

Knoben et al. (2019) is also directly relevant to the cross-location comparability point Reviewer 1 raised in comment 1.6, so we now cite it in the benchmark section as well.

2.12 Bias versus error

(L98–99) “The bias of a single forecast point is the difference between observed and forecasted values” defines error (sometimes called residual), not bias. Bias is a systematic tendency.

Suggest correcting the definition.

Corrected, and this was a straightforward error on our part. The text now reads:

“The error, or residual, of a single forecast is the difference between the forecast and the observation. Bias is the systematic component of error, the tendency of a forecast to be too high or too low on average, and is conventionally expressed for a set of forecasts as a percentage of total observed volume.”

We note that the original definition also had the sign backwards relative to the PBIAS equation that follows it, which is written as $\sum(f_i - o_i)/\sum o_i$. The revised wording is consistent with the equation.

2.13 Pearson correlation formula

(Eq. 3) The Pearson correlation coefficient formula is missing a summation over the second term of denominator.

Corrected, and thank you for checking the equations. The denominator was written as a single summation over the product of squared deviations, which is not the Pearson coefficient. It now reads as the product of two separate sums:

$$r = \frac{\sum(o_i - \mu_o)(f_i - \mu_f)}{\sqrt{\sum(o_i - \mu_o)^2 \sum(f_i - \mu_f)^2}}$$

We have checked the remaining equations. The KGE and PBIAS expressions are correct as published. The reported correlation values were computed with a standard library function and are unaffected by the typesetting error.

2.14 Nominal capacity

(L107) 589 megawatts of “nominal” hydropower capacity. Is this installed capacity or average generation? Please clarify.

It is installed nameplate capacity, not average generation. “Nominal” was ambiguous and we have replaced it. The text now reads “589 megawatts of installed nameplate hydropower capacity across 13 generating stations on the Connecticut and Deerfield Rivers in ISO New England,” which also makes clear that the system total covers many more facilities than the three evaluated here.

2.15 Relationship among the three sites

(L109) Add a sentence describing how these 3 sites are related, given they are downstream from one another one could assume they are operated jointly. Also good to note why you chose these sites over all the other sites (green dots in the figure).

Agreed, and this omission made the results harder to interpret than necessary. Your inference is correct. The three dams lie in series on the Connecticut River mainstem, Wilder upstream, then Bellows Falls, then Vernon, and they are operated jointly, with releases from each becoming inflow to the next. This is why the drainage areas are nested and why bias propagates downstream, an effect we noted in the results without having given the reader the information needed to understand it. We now quantify the propagation (see comment 1.5).

On site selection, these three were chosen because they are the largest generators in the system, they are the dams subject to the FERC forebay constraint that motivated the tailored metrics, and they are where the operators concentrate their scheduling attention. The three tributaries were included because they are the largest unregulated inputs between the dams. The remaining facilities in Figure 2 are smaller and are not covered by the same license condition. Both points are now stated in the text, along with the drainage area table added for Reviewer 1.

2.16 Anonymization and reproducibility

(L117) Anonymising the commercial forecast products is understandable but limits reproducibility. Consider at minimum describing the general class of model used by each (e.g., physics-based, statistical, machine learning) so readers can contextualise the results.

Agreed. Reviewers 1 and 4 asked for the same thing, and all three of you identified the same deficiency. Forecast A is a conceptual rainfall-runoff model with hydraulic routing driven by commercial numerical weather prediction, issued daily; forecast B is a machine learning model trained on gauge and remotely sensed observations, issued twice daily; the in-house forecast combines RFC forecasts with routing and operator judgment. Please see our response to comment 1.4 for the full text and for what we are not able to disclose.

We have also added the inability to inspect commercial products to the limitations, since it constrains any evaluation of this kind and is part of what motivates judging products on operational performance.

2.17 SI units

(Figure 1) Y-axis units are “cfs” (cubic feet per second). Consider also providing SI units for international readers.

Added. The Figure 1 caption now gives the conversion ($1,000 \text{ cfs} \approx 28.3 \text{ m}^3/\text{s}$), and we give SI equivalents at first use in the text: the 0.5 foot forebay limit is noted as approximately 15 cm, as you observed, and the largest observed inflow is given in both units. We have kept cfs as the primary unit because it is the unit the operators and the FERC license use, and the tailored metrics are expressed in feet of forebay for the same reason, but the conversions are now available. We have not added a secondary axis to the figure itself; if the editor prefers that we will do so.

2.18 Seasonal performance

(L134) “Forecasts tend to perform worse in the winter which is the wet season and worse in the summer which is the dry season.” This says forecasts perform worse in both wet and dry seasons. Please clarify when forecasts actually perform well and why. The following sentence about snow melt season partially addresses this but the logic is unclear.

You are right that the sentence is self-contradicting as written, and the logic was garbled. In re-examining the monthly results to answer this we also found that the original characterization of the seasons was wrong on the facts: winter is not the wet season in this basin. Mean inflow at Vernon is highest in March and April (25,500 and 27,100 cfs) and lowest in September through November (roughly 6,000 cfs), so the high-flow season is spring, driven by snowmelt, and that is also where forecast skill is highest. We have rewritten the passage:

“Skill is highest during the spring snowmelt period, roughly March through May, for all three

products. It is lowest in late summer and fall for the commercial products, and the sharpest degradation of any product in any season is forecast A in January and February, where pooled day-ahead KGE falls to 0.39 and -0.40 respectively. The seasons where performance is poorest are those in which inflow is driven by discrete precipitation events: rain and rain-on-snow in winter and convective storms in summer. Both depend on precipitation timing and intensity at scales that are difficult to predict. Snowmelt-driven inflow is comparatively predictable because it is governed by an accumulated snowpack that can be observed in advance and released over a period of weeks, so skill depends more on temperature, which is forecast more reliably than precipitation.”

The original text also stated that the snowmelt season is April through July, which does not match the monthly inflow pattern; it is now given as March through May.

2.19 The inflow-equals-outflow assumption

(L146) The 0.5 foot (≈ 15 cm) operating range is an important constraint that motivates a strong tailored metric. However, the forebay exceedance analysis assumes an inflow=outflow condition which is operationally unrealistic as the authors note. It would strengthen the analysis to discuss how operators actually deviate from this assumption and what the real-world exceedance rates might be.

A fair point, and we have expanded the discussion of what the metric does and does not represent. Operators do not hold pass-inflow conditions: they schedule generation to a market bid, which means departing from inflow during the day and recovering the forebay overnight, and they intervene when the forebay approaches the band by adjusting generation or spilling. The pass-inflow assumption therefore describes what would happen if the forecast were used without intervention.

We think this is the right quantity for comparing forecast products, and we now say why rather than merely flagging the assumption as unrealistic:

“The metric therefore describes what would happen if the forecast were used without intervention, which makes it a measure of the compliance risk a forecast carries absent mitigation rather than a prediction of observed exceedance rates. We consider this the right quantity for comparing forecast products, because it isolates forecast quality from the operators’ skill at correcting for it. Realized exceedance is less frequent than these probabilities imply, but realized rates reflect both the forecast and the effort spent compensating for it, and that effort is itself a cost of a poorer forecast.”

We have not attempted to estimate real-world exceedance rates. Doing so credibly requires representing the intervention behavior, which returns to the scheduling model coupling discussed under comment 4.9, and would not be defensible without it. We now state this as a limitation rather than leaving the assumption unexamined.

2.20 Extremes and the record length

(L178–182) The discussion on limitations of real-time vs hindcast evaluation is good. Consider also noting the limitation that almost two years of data may not capture the full range of extreme events which are often the conditions where forecast quality matters most for operational decisions.

Agreed, and your point that extremes are where forecast quality matters most is worth stating

explicitly, which the original discussion did not. Reviewer 4 asked in comment 4.2 what the period did contain, so we have addressed both together, and checking the second question changed our answer to the first.

The period captured the July 2023 Vermont flood, which produced the largest calculated inflows of the evaluation at four of our six locations, and a December 2023 rain-on-snow event that produced the largest inflow at Wilder. Checking these against the long-term annual peak series at the USGS reference gages places them in context: the July 2023 event ranks among the ten largest flows on record on the Connecticut River main stem, but below the 1936, 1938, and 1927 floods and below the 2011 Tropical Storm Irene peak. The period also did not include a severe multi-year drought. So the record covers a useful range but not the upper tail, and we now report both the events observed and where they sit in the historical distribution. We have adopted your framing about why the limitation matters, since extremes are the conditions under which forecast quality most affects operational decisions. Please see comment 4.2 for the full text.

Reviewer 3

However, proposed framework itself is not well described, such as what is new, different from existing ones, and comparison of past studies are insufficient.

Thank you for this review. Your observation that we implied the framework was novel without describing the novelty was the most useful single comment we received, and the handling editor highlighted it as well. Our response to the editor above describes the new statement of contribution. Your comments 3.4 and 3.5, asking for a comparison against traditional methods and for a case where the tailored metrics change the conclusion, are addressed together below and have produced what we now consider the central result of the paper.

3.1 “market prices”

(l47) “market prices” — *Could you please clarify what it means?*

The sentence was too compressed. The point concerns why a hindcast evaluation is difficult for a hydropower operator: reconstructing what a forecaster would have done in a past period requires knowing the conditions they faced, and those conditions include the wholesale electricity prices they were bidding against, which drive the generation schedule and hence the release decisions. Those price signals, and the operator’s response to them, are not generally recoverable after the fact. We have rewritten the passage:

“[W]here the forecast involves human judgment, as the in-house forecast evaluated here does, a hindcast requires forecasters to reconstruct the decisions they would have made under past conditions. Those conditions include the wholesale electricity prices the operator was bidding against, which drive generation scheduling and therefore release decisions, along with the state of the data feeds at the time. Reconstructing these after the fact is difficult and the reconstruction is difficult to validate.”

Reviewer 2’s comment 2.6 prompted a further correction to this same passage: the claim that a hindcast requires re-operating the system was overstated for a quality assessment, and the requirement properly applies to a value assessment. The revised text separates the two.

3.2 How feedback is reflected in the process

(150, §2.1 real time inflow forecast evaluation) Step 1 through 3 are so general. I assume that steps 4 and 5 are unique points in this manuscript. Please provide a more detailed explanation of how feedback is reflected in step 4.

You have identified the structure of the contribution correctly, and we have rewritten the section to say so directly rather than presenting all the steps as equally novel. Steps 1 through 3 are generic and we now label them as such, claiming no novelty for them.

Your comment also exposed an ambiguity in how we had written the later steps: what was a single step for soliciting feedback conflated gathering operator comment with actually changing the metric set. These are different activities and the second is the one that distinguishes the framework, so we have split them into separate steps, “Gather feedback” and “Update metrics.” The process is now described in six steps, with steps 4 through 6 being where it departs from retrospective verification.

We have expanded both with the mechanics of how feedback was actually collected and how it changed the evaluation:

“Feedback was collected in recurring meetings with the operators, held monthly for most of the evaluation. Each meeting reviewed the current calculated metrics against recent operating experience, and operators were asked to identify which recent forecast errors mattered operationally, and which did not. The answers frequently diverged from what the generic metrics indicated, especially during the high-flow season. Operators discounted large errors during high-flow spill conditions, when they had little discretion over releases and forecast quality was operationally irrelevant, while treating moderate errors with the potential to impact the forebay compliance band as serious. These ongoing discussions motivated the tailored metrics: the forebay exceedance metrics were designed in direct response to operators stating that compliance risk, rather than average error, determined whether they trusted a forecast.”

We have also added a table tracing each tailored metric to the operator concern that prompted it and the approximate point in the evaluation at which it was added, so the process is auditable rather than asserted:

Metric	Operator concern	Added
Day-ahead volumetric skill	Day-ahead market bids are set from next-day inflow volume, not hourly shape	Months 1–2
Seasonal metric breakdown	Products were suspected to behave differently in snowmelt and rain driven regimes	Months 3–4
Cumulative forebay error	Average error does not indicate whether the forebay band would be breached	Months 4–6
Forebay exceedance from a non-neutral start	Operators rarely begin a day at mid-band, which is the harder and more common case	Months 5–6

We should be candid that the months given are approximate. The evaluation predates this paper and we did not keep a formal log of metric provenance, so these are reconstructed from meeting materials and correspondence. We think the table is worth including because the sequence and the motivations are accurate and they make the process concrete, but we have described the timing as approximate rather than implying a precision we cannot support. A practical recommendation

follows from this, which we now state: an evaluation of this kind should log metric provenance as it goes, since the record of why each metric exists is part of what makes the results defensible to the operators later.

3.3 Is a one-year learning period required?

(l62, “5. Iterate”) This implies that at least one year learning period is required. Is this understanding correct?

Your understanding is correct for a full evaluation, and your question identified that the original wording conflated two different durations. We have separated them:

“The iteration in step 6 involves two timescales that should not be conflated. Metric development converged within roughly the first six months of our evaluation, after which the iteration served to accumulate sample size rather than to identify new metrics. A minimum defensible skill assessment requires at least one full annual cycle, since forecast performance may vary substantially by season and different forecast models may rank differently in each season. Operators should therefore expect the set of metrics to stabilize well before the evaluation period ends.”

Our own results bear on the second point. Pooled day-ahead KGE for forecast A ranges from -0.40 in February to 0.88 in May, so an evaluation conducted in spring alone would have reached a substantially more favorable conclusion about that product than the full record supports. We have added the tension the operators raised, that industry preference runs toward shorter evaluations, sometimes a single season, while a single season samples only one hydrologic outcome. The framework does not resolve this, and we now say so, recommending that a real-time evaluation be paired with a retrospective check on the longer record wherever one is feasible.

3.4 Comparison with traditional methods

(l105, result of case study) Were all the results derived through the proposed framework? If so, please compare those result with those obtained via “traditional” methods. This conmparison may enable to enhance the effectiveness of the proposed framework.

All results were produced through the framework, and we agree the comparison you describe is what demonstrates its effectiveness. We have added it in two forms.

First, a discussion subsection, “Comparison with retrospective forecast evaluation,” setting out what each approach can and cannot deliver. A retrospective evaluation provides larger sample sizes, a wider range of hydrologic conditions, and better coverage of extremes, since hindcasts can span decades where our evaluation spans less than two years. Against that, translating thousands of historical forecasts into operational decisions may be prohibitively expensive, and even where historical operations could be reconstructed it is difficult or impossible to reconstruct the exact market conditions at the time of a decision. A real-time evaluation reflects actual operational conditions including outages and market feedbacks, and requires no additional data or extra operator effort. We are explicit that the real-time framework is not a replacement for retrospective verification where a hindcast is feasible; it is a practical alternative where one is not.

Second, and more directly, the comparison of conclusions. A conventional verification of these forecasts, computing NSE, KGE, PBIAS, and correlation over the full record, concludes that forecast B outperforms forecast A and is comparable to the in-house forecast. That conclusion is correct

and this framework reproduces it. What the conventional approach omits is the compliance risk finding that carried the most weight with the operators, detailed under comment 3.5 below. We are careful there not to overstate the point: the two approaches agree on the ranking, and what the tailored metric adds is a dimension of comparison the generic metrics leave invisible rather than a contradiction of them.

3.5 Advantage of tailored metrics

(l139, §3.1 Tailored metrics) Please explain the advantage of introducing tailored metrics. For example, it would be helpful to show a case where using tailored metrics led to a different forecast conclusion compared to general metrics.

This was the right thing to ask for and we have added exactly that case. It is now presented as a result rather than left for the reader to infer.

We should be precise about the strength of the claim, because our own results do not support the stronger version. The generic metrics and the tailored metrics do not reach opposite conclusions about which product is better. Forecast B outperforms forecast A on the generic metrics and also on the tailored metrics, so the overall ranking is consistent. What the tailored metric does is separate the products in a dimension the generic metrics leave invisible, and quantify the separation in units the operators are accountable for.

The forebay exceedance metric is that case. From a neutral forebay starting position, the probability of breaching the compliance band is below 10% for every product at every lead time in the 1–3 day range, and on that basis the products are difficult to distinguish. From the non-neutral starting position that operators are routinely in, the products separate sharply: at Bellows Falls forecast A reaches 23.7% on the first day and 11.9% on the second, against 1.8% and 2.4% for forecast B; at Vernon forecast A reaches 10.1% by day 3 against 5.2%. A roughly tenfold difference in compliance risk at Bellows Falls is not visible anywhere in the NSE, KGE, PBIAS, or correlation values, all of which weight every timestep by error magnitude alone and none of which know that a penalty attaches to a particular forebay threshold.

The point worth emphasizing is that the non-neutral case exists in the analysis only because the operators asked for it. The neutral-start case, which is what a analyst would naturally compute and what our original draft emphasized, shows nothing. This is the mechanism by which operator participation in metric design changes the result rather than merely improving its reception.

We have added the general point to the manuscript:

“The general point we would like to convey in this section is the importance of computing an array of traditional and tailored metrics. Hydrologic forecasts are highly complex and nuanced, and rarely will there be a cut and dry answer to which is better. Tailored metrics provide information to operators in ways that might be obscured by traditional evaluation metrics, ultimately providing more robust information for investment decisions.”

On comparison to past studies, the introduction already discussed the head-to-head evaluation exercises (the HEPEX community experiments and the Bureau of Reclamation forecast rodeos) and we have sharpened the statement of what distinguishes this work: those exercises evaluate forecast quality in general, while this framework evaluates fitness for one utility’s operating constraints under live conditions.

Reviewer 4

The overall idea is relevant and practically useful for hydropower operators seeking to assess competing forecast products. The paper is generally well-written, and the iterative, operator-inclusive evaluation approach adds value.

Thank you for the careful reading and for the specific, actionable comments. We have addressed all nine. Your comment 4.2 in particular led us to check the evaluation period against the long-term record, which corrected a claim we would otherwise have overstated.

4.1 Smoothing window size and sensitivity

Lines 80–81 recommend a triangular smoother for calculated inflows but do not specify the smoothing window size (number of days or timesteps). This is a critical parameter because the choice of window directly affects computed evaluation metrics. The window size must be stated explicitly, and a sensitivity analysis or justification for the chosen value would strengthen the paper.

You are right that this parameter was left unstated and that it affects the metrics. We have added both the value and a sensitivity analysis. Reviewer 2 raised the same point in comment 2.8.

The smoothing is a 24-hour triangular window on hourly calculated inflow, implemented as two successive 24-hour moving averages, which is what produces the triangular weighting. The window was chosen to match the operational timescale: the day-ahead volumetric forecast is the quantity of interest, so smoothing at 24 hours removes sub-daily forebay noise without attenuating the daily signal being evaluated.

We repeated the day-ahead evaluation at window widths of 0 (no smoothing), 6, 12, and 24 hours. Day-ahead NSE at each dam:

Location	Forecast	0 h	6 h	12 h	24 h
Wilder	A	0.70	0.72	0.72	0.72
	B	0.82	0.83	0.83	0.83
	GRH	0.87	0.87	0.88	0.88
Bellows Falls	A	0.50	0.51	0.52	0.55
	B	0.90	0.90	0.90	0.91
	GRH	0.84	0.83	0.84	0.85
Vernon	A	0.79	0.79	0.79	0.81
	B	0.88	0.88	0.88	0.89
	GRH	0.88	0.88	0.88	0.89

Metrics improve modestly with window width, as expected since smoothing removes observation noise that no forecast can reproduce. The largest change is 0.05 NSE (Bellows Falls, forecast A, 0 to 24 hours). The improvement is not strictly monotonic in every case: the in-house forecast at Bellows Falls dips from 0.84 to 0.83 between 0 and 6 hours before recovering, which we attribute to sampling rather than to any property of the smoother. Critically, the ranking of the three products is unchanged at every window width at every location, and the gap between the two commercial products is stable: forecast B leads forecast A by 0.39 NSE at Bellows Falls with no smoothing and by 0.37 with 24-hour smoothing. The conclusions of the evaluation do not depend on this choice. The 24-hour column reproduces Table 1 exactly, so the sensitivity analysis and the main results

come from the same pipeline.

We have added the table and the following caution, which we think is the more generally useful part of the answer:

“Evaluators should nonetheless state the window explicitly and check that their conclusions are insensitive to it, since a window wide enough to attenuate the signal of interest would flatter forecasts that under-predict variability. We recommend matching the window to the decision timescale rather than choosing the window that maximizes metric values.”

4.2 Hydrologic representativeness of the evaluation period

Line 116 states the evaluation ran from July 2023 to May 2025. The authors should discuss whether this period captured any notable extreme hydrologic events (major floods, drought conditions) and how climatologically representative it is relative to the long-term record for the Connecticut River basin. This context is important for understanding the generalizability of the approach.

This was a valuable comment and answering it properly changed two things in the manuscript.

First, the stated period was wrong. The record actually analyzed runs from June 2023 to April 2025, not July 2023 to May 2025, and more importantly the three products do not cover it uniformly: forecast B spans the full period (657 day-ahead forecasts per dam) while forecast A and the in-house forecast both end in November 2024 (539 and 528 respectively). We have corrected the period, added the per-product sample sizes to Table 1, and verified that restricting the comparison to the 510 days on which all three products are present leaves the ranking unchanged at all three dams, with individual metrics shifting by no more than 0.03 NSE. We now report this partial coverage as a characteristic feature of real-time evaluation rather than smoothing over it.

Second, on the events themselves. The period captured the July 2023 Vermont flood, which produced the largest calculated inflows of the evaluation at Vernon (83,100 cfs, 2,353 m³/s) and Bellows Falls (67,000 cfs), and a December 2023 rain-on-snow event that produced the largest inflow at Wilder (27,600 cfs) and the second largest at Vernon and Bellows Falls. It also included spring freshets in 2024 and 2025 and dry summer conditions in both 2023 and 2024. We checked the July 2023 event against the long-term annual peak series at the USGS reference gages to place it in context: it ranks among the ten largest flows on record on the Connecticut River main stem, though below the 1936, 1938, and 1927 floods and below the 2011 Tropical Storm Irene peak. So the period was not hydrologically quiet, but neither did it sample the upper tail. We have written both findings into the manuscript:

“The period therefore sampled a reasonable range of the conditions relevant to GRH operations. It did not, however, sample the full distribution of hydrologic conditions. Notably, the July 2023 event produced one of the ten largest flows ever recorded on the main stem of the Connecticut River. The evaluation also did not include any severe multi-year drought. Any real-time evaluation is bounded in this way, and the conclusions below should be taken as conditional on the observed period.”

Reviewer 2 asked in comment 2.20 for the limitation to be stated explicitly, along with the observation that extremes are where forecast quality matters most, and we have added that to the limitations discussion. One caveat on the comparison itself is worth noting: the gage annual peak series are instantaneous peak discharge at the gage, while our values are 24-hour smoothed calculated inflow at the dams, so the two are not strictly commensurable. We have therefore phrased

the manuscript statement as an approximate ranking rather than a numerical comparison.

4.3 Characterization of the commercial forecasts

Line 117 The authors anonymize the two commercial forecasts as A and B. While commercial confidentiality is understandable, the paper should provide at least a general characterization of each forecast’s methodology (e.g., physics-based, machine-learning-based, hybrid, statistical). Without this, readers cannot interpret the results in a broader methodological context or assess the generalizability of the findings.

Agreed, and Reviewers 1 and 2 raised the same point. Please see our response to comment 1.4 for the characterization we have added and for what we are not able to disclose. In summary: forecast A is a conceptual rainfall-runoff model with hydraulic routing driven by commercial numerical weather prediction, issued once daily out to roughly 8 days; forecast B is a machine learning model trained on gauge and remotely sensed observations, issued twice daily out to 10 days; the in-house forecast combines RFC forecasts with routing and operator judgment and extends about 32 hours. The issue frequencies and horizons are taken from the forecast archive rather than vendor documentation.

We have also added the observation that neither vendor released model internals or calibration diagnostics, which is relevant to the framework’s motivation rather than merely a caveat.

4.4 Extension to probabilistic and ensemble forecasts

The paper focuses only on deterministic forecast evaluation (lines 42–49). While this is acknowledged, the framework could be briefly extended to probabilistic or ensemble forecasts using metrics such as CRPS, reliability diagrams, or rank histograms. This would make the description more complete and forward-looking, especially given the authors’ own statement that probabilistic forecasts are gaining in popularity.

We agree this makes the framework description more complete and have added a methods subsection. Reviewer 2 asked for the same in comment 2.5. The framework’s structure is agnostic to whether the forecast is deterministic or probabilistic, since what changes is the metric set in step 2, not the process. The new text covers the continuous ranked probability score as the probabilistic analogue of the error metrics we recommend, noting that it reduces to mean absolute error for a deterministic forecast so the two can be compared on a common scale, and that it decomposes into reliability and resolution in the same diagnostic spirit as the KGE decomposition; rank histograms for diagnosing ensemble spread; and reliability diagrams for threshold exceedance forecasts, which map directly onto the tailored metrics we developed.

The connection to our own tailored metric is the part we found most worth making:

“The forebay exceedance metric presented there is in effect a probabilistic statement derived from a deterministic forecast, computed across the empirical distribution of that forecast’s past errors. Given an ensemble it would be computed directly from the ensemble members, which is both simpler and better founded.”

We retain the operational caveat, since it is why the deterministic case remains relevant:

“Operators bidding into a day-ahead market must submit a single quantity, so a probabilistic forecast must be reduced to a point value at the moment of the bid. Probabilistic information is valuable for assessing the risk around that value, particularly for compliance limits of the

kind evaluated here, but improvements in deterministic skill remain directly useful. An evaluation of a probabilistic product for this application should therefore assess both the full predictive distribution and the point forecast derived from it.”

This is a description of how the framework would extend rather than a demonstration, since all three products evaluated here are deterministic and we have no ensemble product to apply it to.

4.5 Hyphenation of “real time”

(Introduction) Throughout the manuscript: “real time” should be hyphenated when used as a compound adjective (e.g., “real-time evaluation”). Usage is inconsistent.

Corrected throughout, including the title. Reviewer 2 raised the same point in comment 2.3, where we describe the treatment in detail. In brief: the manuscript contained 29 instances of “real time” and none hyphenated, and we reviewed each individually, hyphenating in attributive position and leaving the adverbial phrase (“conducted in real time”) unhyphenated.

4.6 KGE citation

(Methods, line 92) Remove the placeholder “(refs)” and insert the actual KGE citation (Gupta et al. 2009).

Corrected. The placeholder is replaced with Gupta et al. (2009), and we have added Nash and Sutcliffe (1970) for NSE in the same sentence. We also corrected the misspelling Reviewer 2 caught in comment 2.9: the manuscript read “King-Gupta Efficiency” and now reads “Kling-Gupta Efficiency.”

While checking this we found and fixed a broken citation key elsewhere in the manuscript: the introduction cited `kratzertLearningUniversalRegional2019` where the bibliography entry is `kratzertLearningUniversalRegional2019a`, so that citation was not resolving. We have verified that all citation keys in the manuscript now resolve against the bibliography.

4.7 RMSE or MAE in Table 1

(Results and Case Study, Table 1) Consider adding RMSE or MAE values alongside the existing metrics to provide a sense of forecast error magnitude in physical units.

Agreed, and this was a real omission since every metric in the original table was dimensionless, leaving readers without any sense of error magnitude. We have added RMSE in cfs to Table 1. The values are informative in themselves: forecast A’s day-ahead RMSE at Bellows Falls is 5,904 cfs against 2,615 for forecast B, which conveys the practical size of the difference more directly than the NSE values of 0.55 and 0.91. Table 1 also gains the KGE skill score column described under comment 1.6 and the per-product sample size described under comment 4.2.

4.8 Figure 5 caption

(Figure 5 caption) The caption does not specify that the error bars represent the 2.5th and 97.5th percentiles (as stated in the main text at lines 148–149). Figure caption needs to be self-contained.

Corrected. The Figure 5 caption now states that points show the median across all forecasts and error bars span the 2.5th to 97.5th percentile, and specifies the pass-inflow assumption and the

± 0.5 foot compliance band. We reviewed all captions for self-containment as part of the caption revision described under comment 1.8.

4.9 What coupling to scheduling models would entail

(Discussion, lines 173–174) The authors note that coupling this evaluation with hydropower scheduling models is an important next step. This point could be strengthened by briefly discussing what such a coupling would entail and what data would be needed.

Agreed, the original text asserted the next step without describing it. We have expanded the passage to state what the coupling requires:

“Such a coupling would route each candidate forecast through a scheduling model to produce the generation schedule it implies, then value that schedule against realized inflows and market prices, so that forecast differences are expressed in revenue rather than in skill metrics. This requires the utility’s scheduling model with its operating constraints and unit efficiency curves, historical day-ahead and real-time market prices at the relevant node, the realized inflow record, and the archive of forecasts as issued so that schedules are constructed only from information available at bid time. That last requirement is a further reason a real-time evaluation is a useful precursor, since it produces exactly this archive as a byproduct.

The difficulty is that scheduling models are typically configured to reproduce a utility’s current bidding strategy, which was itself developed around the accuracy of its existing forecast. A more accurate forecast may show little revenue benefit simply because the model is not set up to exploit it. Quantifying the value of improved forecasts may therefore require revising the scheduling model and the operating strategy alongside the forecast, which makes this a substantially larger undertaking than a metric calculation.”

This connects to Reviewer 2’s first major comment, since the same coupling is what a full forecast value assessment would require, and to comment 2.19, since estimating realized forebay exceedance rates rather than unmitigated compliance risk needs the same machinery.

Summary of changes

Framing and positioning

- New statement of contribution at the end of the introduction, distinguishing the framework from retrospective verification on two specific grounds: operator participation in metric selection and design, and evaluation on the operational data feed (editor, 1.1, 2.2, 3 general).
- Introduction gap statement rewritten so that the gap is the absence of an evaluation procedure. The new text states why traditional verification is often infeasible (hindcast cost and data requirements) and why evaluating archived operational forecasts is a poor substitute for continuously updated, operator-adjusted, or third-party products (2.4).
- Process restructured from five steps to six, separating “Gather feedback” from “Update metrics” since collecting operator comment and changing the metric set are distinct activities. Steps 1–3 explicitly labeled as generic practice, with novelty claimed only for 4–6 (2.2, 3.2).
- New discussion subsection “Comparison with retrospective forecast evaluation” (3.4).

- New discussion subsection “From forecast quality to forecast value,” drawing the quality–value distinction, citing Farkas et al. (2025) as a systematic review of the field, and giving operators a two-stage pathway from the metrics presented to a value assessment (2.1).
- Abstract revised to state what distinguishes the framework and to reference the case where a tailored metric separated products that the standard metrics could not.

New results and analysis

- New result: the forebay exceedance metric separates the two commercial products where the generic metrics do not. From a neutral forebay start all products are below 10% exceedance and are difficult to distinguish; from the non-neutral start operators are routinely in, forecast A reaches 23.7% at Bellows Falls against 1.8% for forecast B (3.5).
- New diagnosis of forecast A’s low NSE at Bellows Falls: the KGE decomposition shows an almost pure correlation deficit with no variability or bias error, and evaluating the three tributaries individually localizes it to the unregulated local runoff between Wilder and Bellows Falls, where forecast A has day-ahead NSE of -0.02 , 0.08 , and -1.22 (1.6).
- Smoothing window stated explicitly (24-hour triangular) with a new sensitivity table over 0, 6, 12, and 24 hours showing the forecast ranking is unchanged; the 24-hour column reproduces Table 1 exactly (4.1, 2.8).
- KGE skill score against the persistence forecast added as a column to Table 1, with a new equation defining a skill score generally, and text explaining that NSE and KGE are normalized by observed variance and so are not comparable across locations (1.6, 4.7).
- RMSE in cfs added to Table 1 (4.7).
- Per-product sample sizes added to Table 1; common-period check reported, showing the ranking is unchanged (4.2).
- Downstream bias propagation quantified rather than asserted (1.5, 2.15).
- Evaluation period corrected to June 2023–April 2025 and per-product coverage reported (4.2).
- Hydrologic characterization of the evaluation period, including the July 2023 Vermont flood and December 2023 rain-on-snow event, with their position in the long-term annual peak record (4.2, 2.20).
- Seasonal passage corrected: spring is the high-flow season and where skill is highest, not winter; the self-contradiction removed and the mechanism explained (2.18).

New descriptive material

- New subsection on GRH operating context: run-of-river operation, the daily cycle, the day-ahead market, and the FERC forebay limit (1.3).
- New subsection on the forecast products, characterizing all three by model type, forcing, issue frequency, and horizon, with the inability to inspect commercial products noted as a limitation (1.4, 2.16, 4.3).
- New table of drainage areas and USGS reference gages for all six evaluation points, plus basin

description including mainstem nesting and upstream flood control regulation (1.5).

- New table tracing each tailored metric to the operator concern that prompted it and the approximate month it was added (3.2).
- Steps 4 and 5 expanded with the mechanics of feedback collection and how it changed the metric set (3.2).
- Iteration step clarified to separate metric convergence from record length (3.3).
- New methods subsection extending the framework to probabilistic and ensemble forecasts (CRPS, rank histograms, reliability diagrams) with the day-ahead bidding caveat retained (4.4, 2.5).
- Expanded discussion of what coupling to a scheduling model would entail, the data required, and why the current bidding strategy limits measurable benefit (4.9).
- Expanded treatment of the pass-inflow assumption: the metric measures unmitigated compliance risk, realized rates are lower, and estimating them requires the scheduling model coupling (2.19).
- Added that calculated inflow is itself operations-dependent, so the reference data is not independent of operations (2.6).

Corrections

- Pearson correlation equation corrected: the denominator now has separate summations over each squared deviation term (2.13).
- Bias definition corrected to distinguish error/residual from systematic bias, and made consistent in sign with the PBIAS equation (2.12).
- KGE rationale corrected: the advantage is decomposition into correlation, variability, and bias ratios, not insensitivity to large values (2.10).
- “King-Gupta” corrected to “Kling-Gupta”; the “(refs)” placeholder replaced with Gupta et al. (2009) and Nash and Sutcliffe (1970) (2.9, 4.6).
- Knoben et al. (2019) citation moved adjacent to the -0.41 threshold claim it supports, and the consequence stated (2.11).
- Broken citation key `kratzertLearningUniversalRegional2019` fixed; all keys verified against the bibliography (4.6).
- Persistence redescribed as “a simple but competitive benchmark” rather than the worst possible forecast, and its meaningful horizon stated (2.7, 1.2).
- “Nominal” capacity clarified as installed nameplate capacity, with the number of generating stations given (2.14).
- Added that the three dams are in series on the mainstem and operated jointly, and why these six sites were selected (2.15).
- “real time” hyphenated in attributive position throughout, including the title (2.3, 4.5).
- SI units given at first use of cfs and for the 0.5 foot forebay limit, and in the Figure 1 caption

(2.17).

- All figure captions made self-contained, including the Figure 5 percentile definition, the no-skill threshold values, and the qualification on persistence at long lead times (1.8, 4.8, 1.2).
- All tables rebuilt with `booktabs` formatting, consistent significant figures, and units in headers (1.7).
- Farkas et al. (2025) added to the bibliography for the forecast value discussion (2.1).

Requests we were not able to fully satisfy

We would rather name these than let them pass as addressed.

- **Climatological benchmark in Figures 3 and 4** (1.2). We recommend one in the text but have not added one, because a defensible daily climatology of calculated inflow at the dams requires a long record that begins only with the evaluation, and a gage-derived substitute would not be commensurable with the quantity verified.
- **Calibration and validation diagnostics for the commercial products** (1.4). Not released to us by either vendor. We state the gap and treat it as a finding about the procurement situation.
- **Redrawing the figures** (1.8). Captions are substantially revised and we have described what we would change, but we have not regenerated the figures and would welcome specific guidance on which aspects are considered deficient.
- **A full forecast value assessment** (2.1). We cite the literature and give a two-stage pathway but compute no value metric, which would require the utility's damage function and a model of operator response. We have also chosen a review citation over a detailed comparison of specific value metrics, and are glad to expand that treatment if preferred.
- **Realized forebay exceedance rates** (2.19). Requires representing intervention behavior, which needs the scheduling model coupling described under 4.9.
- **Demonstrated probabilistic evaluation** (2.5, 4.4). Described but not demonstrated, as all three products evaluated are deterministic.

We thank the reviewers again for the care they took. The manuscript is materially better for it, and in three places their questions caused us to find and correct errors that would otherwise have gone to print: the stated evaluation period, the seasonal characterization, and the description of the cross-site NSE pattern. Reviewer 1's request that we check the NSE values at Bellows Falls (comment 1.6) turned out to be the most productive comment we received, since following it up located a real deficiency in one of the forecast products that we had not previously identified.